Jiyeon Bae

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DATA VISUALIZATION · HUMAN-AI INTERACTION · AR/VR

EDUCATION

Seoul National University *Sep 2025 – Present*
M.S. in Artificial Intelligence, Interdisciplinary Program in Artificial Intelligence Seoul, South Korea
• Advisor: Prof. Jinwook Seo GPA: 4.0/4.3

Chung-Ang University *Mar 2019 – Aug 2023*
B.S. in Engineering, School of Computer Art Seoul, South Korea
• Summa Cum Laude GPA: 4.32/4.5

RESEARCH EXPERIENCE

Harvard University, Visual Computing Group *Aug 2026 – Present*
Research Intern, Advisor: Prof. Hanspeter Pfister Cambridge, MA, USA
• Developed MyMuse to support post-visit artwork inquiry and reflection, combining retrieval-augmented LLM Q&A with interactive narrative visualization of visitor records [U1]
• Contributing to research on how robot assistants revise or preserve act/ask/wait decisions during collaboration using spoken requests, interaction history, and egocentric video from smart glasses [U3]

Seoul National University, Human-Computer Interaction Lab *Dec 2024 – Present*
Graduate Researcher (2025 – Present); Research Intern (2024 – 2025); Advisor: Prof. Jinwook Seo Seoul, South Korea
• Contributed to Beyond the Feed's framing and coordinated views linking recommendation distributions and interaction history to support exploration and steering [U2]
• Developed a framework and Swift prototypes for Apple Vision Pro, embedding visualizations on physical object surfaces to guide data exploration through container geometry [P1]
• Built four coordinated views comparing artist–era audio profiles via cosine similarity and contrast [P2]
• Developed and evaluated rank-correlation and clustering-based metric selection to reduce bias in dimensionality-reduction evaluation across 96 datasets [C1]

Chung-Ang University, Computer Graphics Lab *Sep 2019 – Feb 2020*
Undergraduate Researcher, Advisor: Prof. Sanghyun Seo Seoul, South Korea
• Built a semi-automated pipeline to reuse images from 3D reconstruction datasets for Faster R-CNN training through bounding-box propagation and annotation-aware augmentation [C2]

PUBLICATIONS

Conference Papers

[C1] **Jiyeon Bae**, Hyeon Jeon, and Jinwook Seo. **Metric Design != Metric Behavior: Improving Metric Selection for the Unbiased Evaluation of Dimensionality Reduction**. IEEE Visualization Conference (IEEE VIS), Short Paper. 2025. [Paper]

[C2] Nahyuk Lee, Hannah Noh, **Jiyeon Bae**, Kyunghun Nam, Sanghyun Seo, and Kyunghyun Yoon. **Development of 3D Reconstruction and Object Recognition Model Based on Single Dataset**. Proceedings of the Korea Multimedia Society Summer Conference, Vol. 23, No. 1. 2020. (in Korean)

Conference Posters

[P1] **Jiyeon Bae**, Mingyu An, Jeongin Park, Seokweon Jung, Kiroong Choe, and Jinwook Seo. **Physical Containers as Framing Conditions for Visualization in Augmented Reality**. IEEE Pacific Visualization Conference (PacificVis). 2026. [Paper] 🏆 **Best Poster Award.**

[P2] **Jiyeon Bae** and Jinwook Seo. **A Multi-Level Visual Analytics Approach to Artist–Era Alignment in Popular Music**. IEEE Pacific Visualization Conference (PacificVis). 2026. [Paper]

Under Review and In Progress

[U1] **Jiyeon Bae**, Chunggi Lee, Jinwook Seo, and Hanspeter Pfister. **MyMuse: A Personal Museum for Continued Inquiry and Reflection**. Under review at ACM CHI 2027.

[U2] Sean Kim, Namhee Kim, **Jiyeon Bae**, Min Hyeong Kim, and Jinwook Seo. **Beyond the Feed: Trajectory-Level Interaction for Understanding and Steering Recommendations Under Short-Term Sensitivity**. Under review at

ACM CHI 2027.

[U3] Chunggi Lee, Florian Zogaj, **Jiyeon Bae**, and Hanspeter Pfister. **Generalizing Multimodal Interaction Decisions to Dynamic Visual Environments in Human–Robot Collaboration** (working title). In Progress. 2026.

Honors and Awards

- **Human Resource Development Program for Industrial Innovation**, KIAT — *Aug 2026 – Jan 2027*
  Funding via SNU R&DB Foundation for a research visit to Harvard University (USD 21,112 plus round-trip airfare).
- **Best Poster Award**, IEEE Pacific Visualization Conference (PacificVis) 2026 [P1] — *Apr 2026*
- **AI Graduate School Scholarship**, Seoul National University — *Fall 2025, Spring 2026*
- **Academic Excellence Scholarship**, Seoul National University — *Fall 2025*
- **Top of the Class Scholarship**, Chung-Ang University — *Fall 2022*
- **Academic Excellence Scholarship**, Chung-Ang University — *Spring 2020, Spring 2021*

Talks and Presentations

- **IEEE Pacific Visualization Conference (PacificVis) 2026** — *Apr 2026*
  Two poster presentations [P1] [P2] — Sydney, Australia
- **IEEE Visualization Conference (IEEE VIS) 2025** — *Nov 2025*
  Oral presentation [C1], Short Papers: Visualization with/for/in AI — Vienna, Austria
- **SNU–CMU HCAI Research Exchange Workshop 2025** — *Oct 2025*
  Poster presentation — Seoul National University, Seoul, South Korea
- **3rd Korea Visualization Workshop (K-VIS 2025)** — *Jul 2025*
  Oral presentation [C1] — POSTECH, Pohang, South Korea

Industry Experience

E4NET Co., Ltd. — *Jul 2022 – Aug 2022*
Intern, Cherry (blockchain-based donation platform)
- Conducted a usability evaluation of the Cherry app, identifying interaction pain points and proposing UI/UX design improvements

Selected Projects

RoleCast: Multi-Agent Feedback for Broadcast Planning — *Apr 2026 – Jun 2026*
- Built a multi-agent system for checklist-based critique and conflict mediation across broadcast production roles
- Grounded feedback in MBC data via Qdrant dense/BM25 retrieval and evaluated it with 5 broadcasting experts

YouTrap: You're Trapped in the YouTube Filter Bubble — *Oct 2025 – Dec 2025*
- Designed a topic visualization system to support awareness of filter bubbles in YouTube recommendations
- Chrome extension built with React/TypeScript and D3 and evaluated in a 12-participant study

Sisyphus: Interactive Media Installation — *May 2022 – Jul 2022*
- Created Maya animations exploring repetitive daily routines and shown to 200+ visitors
- Built a motorized projection-mapping wheel for a Kinect/Unity installation responsive to viewer position

Motion Stimulus Camera — *Oct 2019 – Dec 2019*
- Built a Raspberry Pi camera with physical controls, an LCD, and a film-grain effect
- Designed and 3D-printed its enclosure; ranked first in course project evaluations

Real-Time Facial Motion Capture and AR Avatar — *Oct 2019*
- Integrated a Fuse/Mixamo avatar with real-time facial tracking in Unity AR
- Customized eye joints, skinning, and mouth/tongue blendshapes in Maya

Service

- **Volunteer Programming Mentoring** — *Jun 2019 – Aug 2023*
  Daegu Jeil Community Child Center — Daegu, South Korea
- **Mathematics Tutoring** — *Spring 2022, Fall 2022*
  Korea Student Aid Foundation Tutoring Program, Gunpo Singi Elementary School — Remote